

Top 15 Research Papers

Text-Based Mental Health NLP

A. Foundational depression/mental-state detection

"Predicting Depression via Social Media"

De Choudhury et al. (2013) — The founding paper of this entire field

http://research.microsoft.com/en-us/um/people/horvitz/depression_social_media_icwsm_2013.pdf

"Screening for Depression Using Natural Language Processing: Literature Review"

Teferra et al. (2024)

<https://www.i-jmr.org/2024/1/e55067>

B. Domain-specific language models

Your core NLP backbone

"MentalBERT: Publicly Available Pretrained Language Models for Mental Healthcare"

Ji et al. (2021)

<https://arxiv.org/abs/2110.15621>

"Mental-LLM: Leveraging Large Language Models for Mental Health Prediction via Online Text Data"

Xu et al. (2023)

<https://arxiv.org/abs/2307.14385>

"MentaLLaMA: Interpretable Mental Health Analysis on Social Media with Large Language Models"

Yang et al. (2024)

<https://arxiv.org/abs/2309.13567>

C. Cognitive distortion detection

Directly powers your CBT layer

"Automatic Detection and Classification of Cognitive Distortions in Mental Health Text"

Shickel et al. (2020)

<https://arxiv.org/abs/1909.07502>

"A Survey of Cognitive Distortion Detection and Classification in NLP"

(2025) — Great single reference covering the whole subfield

<https://arxiv.org/pdf/2508.09878>

D. Cognitive reframing / CBT response generation

The core of what makes Zoya "therapeutic," not just a classifier

"Cognitive Reframing of Negative Thoughts through Human-Language Model Interaction"

Sharma et al. (2023, ACL)

<https://arxiv.org/abs/2305.02466>

"Training Models to Generate, Recognize, and Reframe Unhelpful Thoughts"

Maddela et al. (2023)

<https://arxiv.org/abs/2307.02768>

"COCOA: CBT-based Conversational Counseling Agent using Memory Specialized in Cognitive Distortions and Dynamic Prompt"

Lee et al. (2024)

<https://arxiv.org/abs/2402.17546>

E. Empathy modeling

Governs how Zoya's responses feel, not just what they say

"A Computational Approach to Understanding Empathy Expressed in Text-Based Mental Health Support"

Sharma, Miner, Atkins & Althoff (2020, EMNLP)

<https://arxiv.org/abs/2009.08441>

"GoEmotions: A Dataset of Fine-Grained Emotions"

Demszky et al. (2020) — 27-class emotion taxonomy, Google

<https://arxiv.org/abs/2005.00547>

"Towards Empathetic Open-Domain Conversation Models"

Rashkin et al. (2019, ACL) — The EmpatheticDialogues dataset

<https://arxiv.org/abs/1811.00207>

F. Risk detection & safety

For your crisis-escalation logic — non-negotiable for a mental health agent

CLPsych 2019 Shared Task: "Predicting the Degree of Suicide Risk in Reddit Posts"

Zirikly et al. (2019)

<https://aclanthology.org/W19-3003/>

G. Evaluation framework

How the field judges these systems

"Charting the Evolution of AI Mental Health Chatbots from Rule-Based Systems to Large Language Models: A Systematic Review"

Hua et al. (2025, World Psychiatry)

<https://onlinelibrary.wiley.com/doi/10.1002/wps.21352>